

Carbon Dioxide and Global Warming A Story of Coupled Molecular Dynamics

Carbon dioxide (CO_2) is modeled as a linear triatomic molecule:

- Central carbon mass: m
- Two oxygen masses: $M = \frac{4m}{3}$
- Two identical springs of constant k

All motion is constrained along the molecular axis.

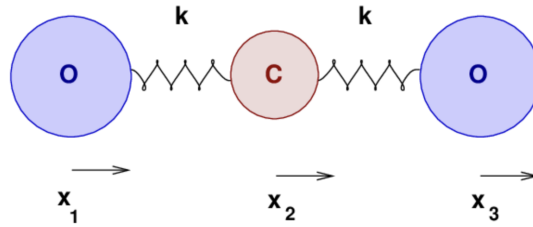


Figure 1: Schematic atomic bonds making up the CO_2 molecule. Credit: RIT

a Let the displacements from equilibrium be

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Masses:

$$m_C = m, \quad m_O = \frac{4m}{3}$$

Equations of motion ($F = ma$):

$$\frac{4m}{3}\ddot{x}_1 = -k(x_1 - x_2)$$

$$m\ddot{x}_2 = -k(x_2 - x_1) - k(x_2 - x_3)$$

$$\frac{4m}{3}\ddot{x}_3 = -k(x_3 - x_2)$$

Divide each equation by its mass to obtain

$$\ddot{\mathbf{X}} = -\mathcal{M}\mathbf{X}$$

where

$$\mathcal{M} = \begin{pmatrix} \frac{3k}{4m} & -\frac{3k}{4m} & 0 \\ -\frac{k}{m} & \frac{2k}{m} & -\frac{k}{m} \\ 0 & -\frac{3k}{4m} & \frac{3k}{4m} \end{pmatrix}$$

- b We expect a total of 3 normal modes since there are three degrees of freedom.
- c Normal modes are special configurations of the N particles where each has the *same frequency* of oscillation. We intend to find these normal mode frequencies ω .

$$\mathbf{X}_{NM} = \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix} \exp(i\omega t)$$

where A_j are complex amplitudes in general. This gives us that the normal modes are eigenvectors and normal frequencies are eigenvalues of the $\mathcal{M}$ matrix,

$$\mathcal{M}\mathbf{A} = \omega^2 \mathbf{A}$$

where

$$= \begin{pmatrix} A_1 \\ A_2 \\ A_3 \end{pmatrix}$$

To get the eigenvalues, we solve the characteristic equation,

$$\det(\mathcal{M} - \lambda \mathcal{I}) = 0$$

Factor out $\frac{k}{m}$:

$$\mathcal{M} = \frac{k}{m} Q,$$

where

$$Q = \begin{pmatrix} \frac{3}{4} & -\frac{3}{4} & 0 \\ -1 & 2 & -1 \\ 0 & -\frac{3}{4} & \frac{3}{4} \end{pmatrix}.$$

Let $\mu = \frac{m}{k} \lambda$. Then

$$\det(\mathcal{M} - \lambda \mathcal{I}) = 0 \iff \det(Q - \mu \mathcal{I}) = 0.$$

So we compute

$$\det \begin{pmatrix} \frac{3}{4} - \mu & -\frac{3}{4} & 0 \\ -1 & 2 - \mu & -1 \\ 0 & -\frac{3}{4} & \frac{3}{4} - \mu \end{pmatrix} = 0.$$

Expand along the first row:

$$= \left(\frac{3}{4} - \mu\right) \det \begin{pmatrix} 2 - \mu & -1 \\ -\frac{3}{4} & \frac{3}{4} - \mu \end{pmatrix} - \left(-\frac{3}{4}\right) \det \begin{pmatrix} -1 & -1 \\ 0 & \frac{3}{4} - \mu \end{pmatrix}.$$

Compute each minor:

$$\begin{aligned} \det \begin{pmatrix} 2 - \mu & -1 \\ -\frac{3}{4} & \frac{3}{4} - \mu \end{pmatrix} &= (2 - \mu)\left(\frac{3}{4} - \mu\right) - \frac{3}{4} \\ &= \mu^2 - \frac{11}{4}\mu + \frac{3}{4} \end{aligned}$$

Second minor:

$$\det \begin{pmatrix} -1 & -1 \\ 0 & \frac{3}{4} - \mu \end{pmatrix} = -1\left(\frac{3}{4} - \mu\right)$$

Putting everything together:

$$\left(\frac{3}{4} - \mu\right) \left(\mu^2 - \frac{11}{4}\mu + \frac{3}{4}\right) - \frac{3}{4}\left(\frac{3}{4} - \mu\right) = 0.$$

Factor out $\left(\frac{3}{4} - \mu\right)$:

$$\left(\frac{3}{4} - \mu\right) \left[\mu^2 - \frac{11}{4}\mu + \frac{3}{4} - \frac{3}{4} \right] = 0.$$

$$\left(\frac{3}{4} - \mu\right) \left[\mu^2 - \frac{11}{4}\mu \right] = 0.$$

$$\left(\frac{3}{4} - \mu\right) \mu \left(\mu - \frac{11}{4} \right) = 0.$$

Thus

$$\mu = 0, \quad \frac{3}{4}, \quad \frac{11}{4}.$$

Recalling $\lambda = \frac{k}{m}\mu$, we obtain

$$\boxed{\lambda_1 = 0}$$

$$\boxed{\lambda_2 = \frac{3k}{4m}}$$

$$\boxed{\lambda_3 = \frac{11k}{4m}}$$

Since $\lambda = \omega^2$, the normal mode frequencies are

$$\boxed{\omega_0^2 = 0}, \quad \boxed{\omega_a^2 = \frac{3k}{4m}}, \quad \boxed{\omega_s^2 = \frac{11k}{4m}}.$$

Putting it back in the eigenvalue equation, we get the eigenvectors (upto overall proportionality)

1. Translational Mode

$$\omega_0^2 = 0, \quad \mathbf{A}_{\text{translational}} = B_1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

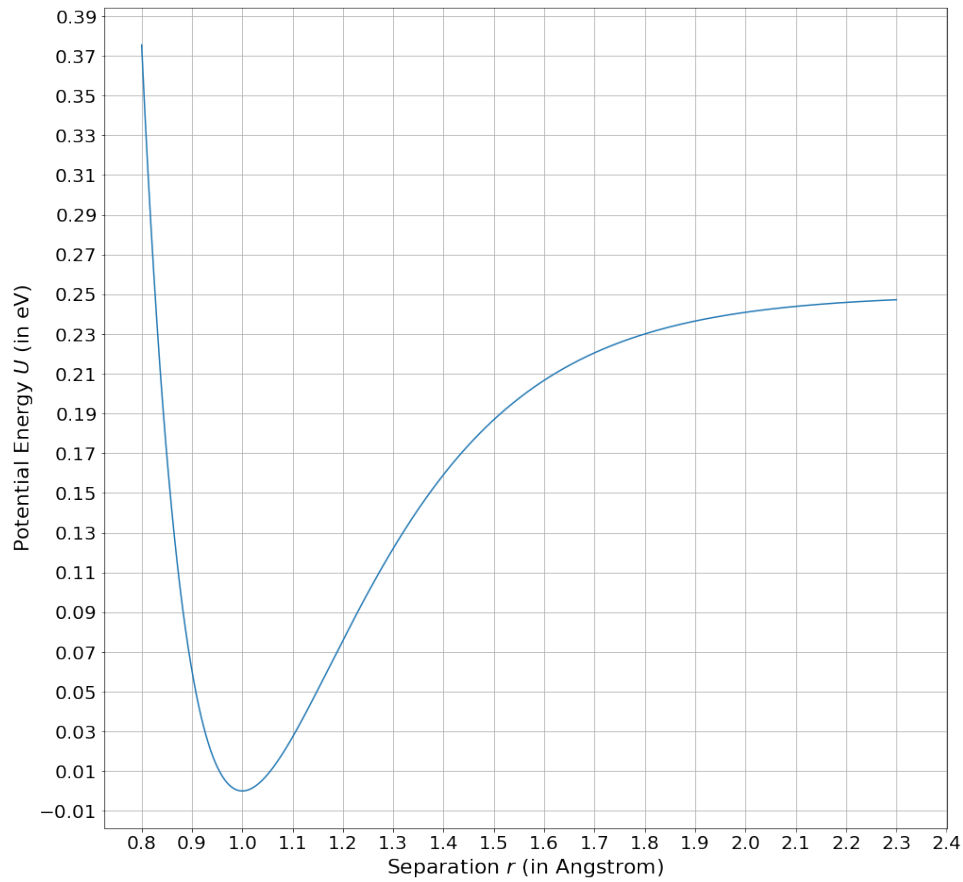
2. Antisymmetric Stretch

$$\omega_a^2 = \frac{3k}{4m}, \quad \mathbf{A}_{\text{anti}} = B_2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

3. Symmetric Stretch

$$\omega_s^2 = \frac{11k}{4m}, \quad \mathbf{A}_{\text{symm}} = B_3 \begin{pmatrix} 1 \\ -\frac{3}{2} \\ 1 \end{pmatrix}.$$

- d The eigenmode with the zero frequency $\omega_0 = 0$ is a translational mode, the entire molecule just moves in a straight line without any oscillation. This is also confirmed by the structure of the eigenvector where each particle has the same amplitude and phase, so simple translation.



e There are actually various ways to get the bond strength/spring constant k .

By estimating all Morse parameter: We are given that the potential is a Morse potential:

$$U(r) = D(1 - e^{-a(r-b)})^2$$

We can look at the graph and infer (for example, seeing the limit as $r \rightarrow \infty$ and the location of the minima $r = b$),

$$D = 0.25 \text{ eV}, \quad b = 1 \text{ Å}$$

Using the point

$$r = 1.8 \text{ Å}, \quad U = 0.23 \text{ eV}$$

$$0.23 = 0.25(1 - e^{-0.8a})^2$$

Solving:

$$a = 4 \text{ Å}^{-1}$$

For a Morse potential, we found in Tutorial 2,

$$k = \left. \frac{d^2U}{dr^2} \right|_{r=b} = 2Da^2$$

Convert:

$$D = 4 \times 10^{-20} \text{ J}$$

$$a = 4 \times 10^{10} \text{ m}^{-1}$$

$$k = 2(4 \times 10^{-20})(16 \times 10^{20})$$

$$\boxed{k = 128 \text{ N/m}}$$

Curvature at Local Minima: Near equilibrium,

$$k = \left. \frac{d^2 U}{dr^2} \right|_{r=b}.$$

Using the central difference approximation,

$$U''(b) \approx \frac{U(b+h) - 2U(b) + U(b-h)}{h^2}.$$

In case you are unsure how this expression came, look at a general function, $f(x \pm h)$ about x using Taylor series:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f''''(x) + \dots$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f''''(x) + \dots$$

Adding the two expansions,

$$f(x+h) + f(x-h) = 2f(x) + h^2 f''(x) + \frac{h^4}{12} f''''(x) + \dots$$

Rearranging gives

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - \frac{h^2}{12} f''''(x) + O(h^4).$$

Thus,

$$\boxed{f''(x) \approx \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}}$$

with truncation error of order $O(h^2)$.

Back to CO_2 , given,

$$b = 1 \text{ \AA}, \quad h = 0.05 \text{ \AA}, \quad U(b) = 0, \quad U(b \pm h) = 0.01 \text{ eV},$$

we obtain

$$k \approx \frac{0.01 + 0.01}{h^2} = \frac{0.02 \text{ eV}}{(0.05 \text{ \AA})^2}.$$

Convert units:

$$0.02 \text{ eV} = 0.02(1.6 \times 10^{-19}) = 3.2 \times 10^{-21} \text{ J},$$

$$0.05 \text{ \AA} = 5 \times 10^{-12} \text{ m}, \quad (5 \times 10^{-12})^2 = 2.5 \times 10^{-23} \text{ m}^2.$$

Therefore,

$$\boxed{k \approx \frac{3.2 \times 10^{-21}}{2.5 \times 10^{-23}} \approx 1.3 \times 10^2 \text{ N/m.}}$$

f Using $k = 128 \text{ N/m}$:

$$\omega_a = \sqrt{\frac{3k}{4m}} \sim 7 \times 10^{13} \text{ rad/s}$$

$$\omega_s = \sqrt{\frac{11k}{4m}} \sim 1.4 \times 10^{14} \text{ rad/s}$$

These lie in the infrared range.

g CO_2 possesses vibrational normal modes in the infrared frequency band, allowing absorption of terrestrial infrared radiation and contributing to the greenhouse effect.
